

Ai categorically has a lower Carbon footprint than people

Posted by u/bro123126 • 0 points • 16 comments

Hello! So this is supposed to be actually argumentative, and I am all for someone changing my opinion, or arguing with what I am presenting.

Just to make one thing clear, I am neither "Pro" Ai or "Anti" Ai, in my honest opinion there is a lot to see in both sides.

I do think AI will revolutionise medicine, science and much more. It can truly become something great, but also there are a bunch of issues, such as people being able to produce deepfakes at home now which is terrifying. And I honestly believe that people who are "Pro" Ai and people who are against it both agree on a lot of things.

The amount of extrem people on both sides does exist but is probably on the lower side. Now to the Main topic, the carbon footprint. I came across a post that mentioned the carbon footprint of AI, and i was interested and looked it up. Here is my main source. [The carbon emissions of writing and illustrating are lower for AI than for humans | Scientific Reports](<https://www.nature.com/articles/s41598-024-54271-x>) I found it pretty trustworthy, and here we have the following statistics for the people who don't want to read the entire thing.

[Please Keep in mind this is ChatGPT 3 Which is more inefficent than 4 thusly it got even better! :)]

([https://preview.redd.it/d41kwbme5i6f1.png?](https://preview.redd.it/d41kwbme5i6f1.png?width=685&format=png&auto=webp&s=e05921b33b24c83d676ab703765492a1aa0da30f)

[width=685&format=png&auto=webp&s=e05921b33b24c83d676ab703765492a1aa0da30f](https://preview.redd.it/d41kwbme5i6f1.png?width=685&format=png&auto=webp&s=e05921b33b24c83d676ab703765492a1aa0da30f))

Here we see the carbon footprint per Image created, unlike what many say, AIs do not use much electricity to create an Image, that's actually one of the Main reasons why we can use those AI models on our Computers. They use very little. [Sam Altman: Energy for ChatGPT Query Can Power Lightbulb for Minutes - Business Insider](<https://www.businessinsider.com/how-much-energy-does-chatgpt-use-average-query-watts-altman-2025-6#:~:text=%22People%20are%20often%20curious%20about,of%20minutes%2C%22%20Altman%20wrote.>) says, that a ChatGPT model, uses 0.34 watts per hour which is basically nothing.

Here is a direct quote:

"People are often curious about how much energy a ChatGPT query uses; the average query uses about 0.34 watt-hours, about what an oven would use in a little over one second, or a high-efficiency lightbulb would use in a couple of minutes," Altman wrote.

"It also uses about 0.000085 gallons of water; roughly one-fifteenth of a teaspoon," he continued...

I think you guys get the point, the environmental impact is truly not so huge as people say.

The entire post stems from a guy who said: "35 Million searches are done on chatgpt..." or something along those lines, which uses a lot of electricity and so on. Which I found very misinformed, because the amount of electricity you are using having your PC on is very similar. Hell, it's more. And saying that we need to regulate the electricity usage of such companies is odd to me, wouldn't that mean that we should regulate people having their PCs on?

Or people using Google search? I mean you get my point, right? If you want to do that then instead of regulating the AI companies you should regulate energy policy and make more renewables or nuclear or whatever clean energy source you like which will minimise the damage on all sides.

Now, I believe there is more on the other side obviously, there is the infrastructure and the training which use significantly more energy, i did find a reliable source for those though, but honestly, if we are able to replace parts of CGI for studios, or use this in medical clinics if it gets good enough or as a search engine and so on, then I'd say it'll be worth it in the long run.

Comments

u/Mataric • 1 points • 2025-06-13 08:21 UTC

It has a much lower footprint than the social media they use to complain about how bad it is for the environment too.

We're talking hundreds of thousands of times worse..

u/MammothPhilosophy192 • 1 points • 2025-06-12 16:47 UTC

that study is posted every once in a while, it's filled with mistakes and assumptions.

search for the other times this has been posted and read the comments.

u/bro123126 • 1 points • 2025-06-12 16:51 UTC

Oh, got it. Can you point out the mistakes for me? it's supported by the National Science Foundation, so I am pretty sure it's accurate, but if there is a mistake I'll make an edit, send me your sources that disapprove this study if you want.

u/MammothPhilosophy192 • 0 points • 2025-06-12 18:42 UTC

honestly I had this discussion too many times, you should look at one of the other times this was posted.

it's counting the co2 of a computer twice, one by itself and once in the carbon footprint of a human.

to correctly measure the consumption you have to study a person on a resting state, then while writing, the delta is the Co2 consumption of writing.

>For this study, we included the hardware and energy used to provide the AI service, but not the software development cycle or the software engineers and other personnel who worked on the AI. Tis choice is analogous to how, with the human writer, we included the footprint of that human's life, but not their parents.

ignoring the training process is absolutely bonkers, you cant use a model without training.

u/Beautiful-Lack-2573 • 1 points • 2025-06-12 15:44 UTC

It's a fun argument to make people mad (not judging), and the environment hysteria has already been debunked so many ways -

but this argument sort of ignores that the human already exists and is eating and breathing regardless.

u/clopticrp • 2 points • 2025-06-12 15:19 UTC

Sweet. Let's get rid of people then and leave it to the AI.

u/bro123126 • 1 points • 2025-06-12 15:31 UTC

\>"Hello! So this is supposed to be actually argumentative, and I am all for someone changing my opinion, or arguing with what I am presenting."

Is this supposed to be an argument? If so then it doesn't relate to the energy policy which was my main topic. As I mentioned in the beginning, this topic is very nuanced. If you want to argue about the energy policy in relation to what I mentioned or sources that you have then please go ahead.

u/clopticrp • 2 points • 2025-06-12 15:46 UTC

It's partially a flippant comment that doesn't do your effort justice, so here goes some more in-depth nuance.

>that a ChatGPT model, uses 0.34 watts per hour which is basically nothing.

Are you mistaking what was said, misquoting it, or do you think that the model uses .34 watts per hour? Read the quote from Altman carefully, .34 *watt-hours per query.* that makes it about 1 watt-hour for 3 queries.

Here's the next bit of nuance. "Ai categorically has a lower Carbon footprint than people".

So what? So does any single thing we use because everything we use makes up our whole footprint. It's a stupid premise to begin with. That doesn't even consider the fact that humanity operates on anthropocentric outlooks. The things - AI, etc, are *for us.* This means that there is no valid comparison *to* us.

One last thing:

> ...because the amount of electricity you are using having your PC on is very similar. Hell, it's more. And saying that we need to regulate the electricity usage of such companies is odd to me, wouldn't that mean that we should regulate people having their PCs on? Or people using Google search?

Prior expenditure or waste does not validate other expenditures or waste. Each thing should be evaluated individually for its utility vs. cost.

u/bro123126 • 1 points • 2025-06-12 16:10 UTC

I am really happy with your reply! And I do apologise for miss quoting, I did actually mean per query, but accidentally wrote per hour for some reason, haha.

Either way! It still doesn't change the fact that this is insignificant. [How Many Watts Does a Computer Use? | EnergySage](<https://www.energysage.com/electricity/house-watts/how-many-watts-does-a-computer-use/>) A pc uses an average of 200 to 500 Watts per hour (For gaming!) and you bet we use way more for CGI and for large studios.

I do find the consideration of the anthropocentric outlook quite interesting, but us being humanity first serves the purpose of supporting AI in my opinion, because it makes research significantly easier and might help with my aforementioned scientific advancements. I would argue that a Movie is a much bigger waste than AI, because there is no real opportunity of any advancement with it other than it being a documentary of course.

"Prior expenditure or waste does not validate other expenditures or waste. Each thing should be evaluated individually for its utility vs. cost."

In theory I agree, that kind of evaluation sounds great. But is it really possible in practice? It's incredibly hard to separate something like "utility" from emotion, culture, or personal values. Think about video games or art. Technically they might not have much traditional utility. They're not curing diseases or solving climate change but that doesn't mean they're not valuable. Most people wouldn't see them as a waste.

If we started judging everything purely based on cost versus utility, we'd end up calling most forms of human creativity and culture pointless, and I don't think that reflects how people actually live or what they care about.

And honestly, if we ever do manage to evaluate things that way, AI might be one of the few technologies that really justifies itself. Its potential in medicine, diagnostics, education, and accessibility is massive.

u/clopticrp • 1 points • 2025-06-12 16:32 UTC

I say it like that because that is how it has to be.

Policy makers, much like AI, cannot hold all of the relevant information necessary in consideration if they start thinking about emotion and personal values. They have to make policy based on numbers.

This means, accurate or not, they will quantify those things - "happiness index", "customer satisfaction score", etc.

That is why, when things start to get difficult, all of the cultural, emotional and personal things get tossed in the bin. When metal meets the grindstone, they just can't be justified by tangible results that outweigh the urgency.

Despite all of this, I think the answer is more investment in energy. One of the very small things I can appreciate coming from Trump is the energy priority and the willingness to seriously expand nuclear energy. I would like to see that balanced with solar.

Those, with distributed storage networks (everyone has a house battery), could deal with most of the issue, I believe, as long as we worked more on the fresh water issue.

Outside of all of that, I have very serious misgivings about the effects of general, public facing chatbots (ChatGPT, Claude, Gemini) on society. I don't think the overall effect is going to be a positive one.

u/bro123126 • 1 points • 2025-06-12 16:47 UTC

"I think the answer is more investment in energy. " Me too! Energy investment is truly one of the most important things in our lives currently. (+ Stability and lower Rents).

Eitherway, I am interested what are your misgivings on chatbots, I know we can't argue about that because it's more subjective than anything, but I am just interested in your opinion. :)

u/clopticrp • 2 points • 2025-06-12 17:09 UTC

Logical train of reasoning:

1. AI chatbots are incapable of knowing when they are wrong. They don't know what a lie is, what feelings are, what truth is, what empathy is. Because of this, they can be confidently wrong *at any given time.*

2. AI chatbots are marketed as "an expert at everything at your fingertips.", despite the fact that they can be, and often are, just wrong, based on #1.

3. Chatbots are designed to sound authoritative, friendly, empathetic, understanding, but it cannot be any of those things.

4. People have a tendency to allow authoritative, friendly, empathetic, understanding

sounding statements to stand without critical examination.

5. People also have "automation bias" or the internalized belief that automation doesn't make mistakes.

6. People listen to the marketing about how good AI is.

7. This means, on average, more people are going to be more wrong, more often because they use AI.

Here's the thing. If you use AI correctly, it actually takes more cognitive effort, not less. You have to critically examine everything it generates for veracity.

This is going to rapidly exacerbate the divide between those who take real advantage of AI to accelerate their productivity, and those who end up spinning their wheels, mired in misunderstandings from a lack of critical thinking. The people in the second group, ironically, will believe they are in the first.

u/bro123126

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u/Minerkillerballer • 1 points • 2025-06-12 14:30 UTC

Antis with environment rhetoric in shamble

u/bro123126 • 1 points • 2025-06-12 14:32 UTC

Haha, I mean I am honestly open to any counter arguments.